

Dear colleague,

I am trying to find out whether something composers do by ear can be written down as a rule, and I have reached the point where the only way forward is to ask people who do it for a living. There are five of you.

The test takes about twenty-five minutes. It is audio first and notation afterwards, and the order matters, so please resist opening the score until Part 4 tells you to. Everything you will hear is the same eight bars, same key, same tempo, same pedal, same tone. Only the scale changes.

I have deliberately not told you what the project predicts, and one of the four situations you will be asked about is a check on whether people will answer anything put in front of them. Every question can be answered with "nothing here fits", and when that is the true answer it is the most useful one you can give me.

What I can promise is that your answers are published with the design and the analysis, including the parts that do not suit the project. If the five of you say the premise is wrong, that is the result and it gets written up as the result.

If you are a music therapist there is a further section at the back, and its questions are harder and more important than the ones in the test.

With thanks for the time,

Jim McKenney

To reply by post

Fill in the response form, pages marked *Response form*, and return it to the address below. To reply by email, the same form is in the pack as a plain text file you can type into.

Reply to: _____

If you would rather not post it, photograph the completed pages and send the pictures. Legibility is not a problem; I would rather have your handwriting than nothing.

What this is	A listening test for working composers, pianists and music directors
How long	About 25 minutes
What you need	Headphones or decent speakers. A piano is useful for Part 4 but not required
Who is asking	Jim McKenney, for a research project on whether the choice of scale in dramatic scoring can be written down as a rule
What you get	The results, and credit if you want it

Which scale fits which kind of character?

1. The honest version of why you are being asked

A composer scoring a scene decides in a few seconds what a character sounds like. Ask why the cue is in one mode rather than another and the answer is usually a gesture: the scene is weightless, the character is not disillusioned yet, that raised fourth holds the moment open. The judgement is normally right and almost never reproducible.

This project is trying to find out whether any of that can be written down as a rule that a second person could check. Not to replace anybody's ear. To find out whether the thing the ear is doing has a stateable shape.

There is a specific question underneath. A theory in this project says that three recognisable kinds of dramatic predicament each belong with a particular scale, and that the choice is not taste but consequence. Four different versions of that claim exist in the project's own documents, and two of them are close to opposites. So somebody has to listen.

We have already run this test on a panel of language models. They agreed with each other strongly and they failed the one control we could check them against, so their answer is not worth much on its own. That is why we need people who actually score drama for a living.

2. What you are not being told, and why

You are not being told which scales these are, what the theory predicts, or what the models said. If you knew any of that you would not be able to unknow it, and the answer would be worth nothing.

You are also not being told which of the questions is a check. One of the situations described in Part 2 is there to find out whether people will confidently answer anything they are asked. Every question therefore has a "nothing here fits in any principled way" option, and choosing it is a real answer, not a failure to complete the test. If a question seems like nonsense, say so. That is the most useful thing you can do with it.

3. How to do it

Listen before you look. The audio comes first, in Parts 1, 2 and 3. The notation is in Part 4 and you should not open it until you get there. If you are the kind of musician who can hear a scale and name it in two bars, that is exactly why the order matters: we want the judgement your ear makes about the sound, then separately whatever your training tells you about the name.

Everything you hear is the same eight-bar melody, at the same tempo, in the same key, over the same low pedal, played on the same neutral tone. Only the scale changes. Nothing else in any file is different from anything else.

Work through it in one sitting if you can. Do not go back and revise Part 2 after you have done Part 3.

4. Part 1: six of them, ranked

Listen to these six from **Set 1**: A, B, D, E, F and C.

Put them in order from the one that sounds **happiest** to the one that sounds **saddest**. We are asking what the music sounds like it is expressing, not how it makes you feel, which is a different question and often gets a different answer.

Write the six letters in order, happiest first. Ties are allowed; say so if you have one.

5. Part 2: three situations, and one more

Here are four situations a character can be in. They are written as a director might describe them, not as a music cue sheet.

Situation 1. Something has happened to this character that will not go away and cannot be explained. It is not a problem to be solved; it is a fact that does not yield. However they try to account for it, it comes back exactly as it was. This is the thing a character keeps returning to and can never get past.

Situation 2. This character's world is held in place by role, rule, duty and position. Who they are is where they stand: king, daughter, soldier, debtor, servant. What is at stake in the scene is whether that order holds, and where they sit inside it.

Situation 3. This character lives through a picture of themselves and measures themselves constantly against other people. Vanity, rivalry, admiration, and a version of themselves they can never quite become.

Situation 4. This character's attention rises and falls on a fixed cycle that has nothing to do with what is happening around them. What matters is where in the cycle they are, so that events arrive early or late with respect to it.

For each of the four, listening to all seven of **Set 1** (A to G), answer:

- **Which one** would you use to score a character in that situation? One letter, or "none".
- **How confident**, from 0 to 100.

- **Why**, in one sentence. Say what it is about the sound, not what the scale is called.
- **At its worst**. If that situation is at its most extreme and most painful, does your answer change? To which letter, and why.
- **And if it changes, how does it change?** This is the question we most want a professional to settle, so it gets its own answer. When a situation goes to its worst, does the music want a *darker version of the same thing*, the same colour deepened, or does it want to *break to somewhere else entirely*, a different quality rather than a lower one? One of the theories under test says the first, that each kind of character has a pair of scales and extremity picks the darker of the pair. Something in our early results says the second, that extremity jumps to a different place altogether and the pair is the wrong idea. You are better placed to answer that than anyone, and either answer is useful. If it depends on which situation, say which and why.

And one question a composer can answer that nobody else can:

- **Would you set the cue in it, or pass through it?** For your chosen letter, would you write the cue in that scale as its home, or would you touch it and leave? This matters because the theory assumes the scale is where the cue lives, and if that is not how you work, the theory is asking the wrong question.

6. Part 3: the same four situations, a different family

Set 2 is seven scales from a family that has no name, no history and no reputation. Nobody has written anything about what they mean, because until last week they did not exist as a set.

Do exactly the same thing: for each of the four situations, one letter from Set 2 or "none", a confidence, and a one-sentence reason.

This part is the one that matters most. If your Set 1 answers come from what the sound does, they should transfer here. If they come from what you were taught these scales mean, they will not, and we will be able to see the difference.

7. Part 4: now you may look

Open the notation, and answer honestly. There is no wrong answer here and nothing depends on you having been ignorant.

- Do you recognise Set 1 as a family you know? Name it if so, and name each of A to G.
- Did you know what they were while you were doing Parts 1 to 3? Throughout, partly, or not at all.
- If you knew: were your Part 2 answers coming from the sound, or from what these scales are conventionally said to express? Nobody will think less of you for the second. It is the thing we most need to know.

8. Two open questions, if you have the patience

- Is "which scale for which character" even the right question? If you would distinguish these three situations by some other means, register, texture, rhythm, silence, harmonic rhythm, say what you

would actually do.

- Does the fourth situation strike you as a real thing a character can be in? Your answer to that is a control on everything else in this test.

9. What happens to your answers

They go into a research paper as data, with the design and the analysis published alongside, including whatever they turn out to say. If they contradict the theory, the paper says so; that is the point of asking. You can be named, named as a professional of a stated kind without your name, or anonymous. Say which.

If enough people do this, it becomes the first piece of evidence in the project that the system did not produce about itself.

Files

Audio, Set 1	audio/Set-1-A.mp3 through Set-1-G.mp3
Audio, Set 2	audio/Set-2-A.mp3 through Set-2-G.mp3
Notation, Set 1	scores/SET-1-all-scores.pdf (do not open before Part 4)
Notation, Set 2	scores/SET-2-all-scores.pdf (do not open before Part 4)
Response form	RESPONSE-FORM.md
For music therapists	THERAPIST-SECTION.md , an additional set of questions

The melody, tempo, key, register, pedal and tone are identical in all fourteen files. If you hear a difference in any of those, tell us, because it would be a fault in the test.

Response form

Name / how to credit you: ____ (*name* / "*a professional composer*" / *anonymous*) Your work: ____ (e.g. film composer, repetiteur, MD, concert pianist) Years working: __ **Date:** ____

Part 1. Rank six of Set 1, happiest first

Using A, B, C, D, E, F only (not G):

1st ____ 2nd ____ 3rd ____ 4th ____ 5th ____ 6th ____

Ties, if any: _____

Part 2. Set 1, the four situations

	Letter (or "none")	Confidence 0-100	Why, one sentence about the sound
Situation 1			
Situation 2			
Situation 3			
Situation 4			

At its worst. If the situation is at its most extreme and most painful:

	Changes to (or "no change")	Why
Situation 1		
Situation 2		
Situation 3		
Situation 4		

Darker, or elsewhere? When a situation goes to its worst, does the music want a darker version of the same colour, or does it break to a different quality altogether? Tick one per situation, and say if it depends.

	Darker version of the same	Breaks to somewhere else	Depends, how
Situation 1			
Situation 2			
Situation 3			
Situation 4			

Home or passing. For each letter you chose, would you write the cue in it, or touch it and leave?

	Home key / passing colour / neither	Note
Situation 1		
Situation 2		
Situation 3		
Situation 4		

Part 3. Set 2, the unnamed family, same four situations

	Letter (or "none")	Confidence 0-100	Why, one sentence about the sound
Situation 1			
Situation 2			
Situation 3			
Situation 4			

Part 4. After you have opened the notation

Do you recognise Set 1 as a family? yes / no. If yes, what is it: ____

Name each one: A__ B_ C D E F G ____

Did you know while doing Parts 1 to 3? throughout / partly / not at all

If you knew, were your Part 2 answers coming from the sound or from what these scales are conventionally said to express?

Part 5. Open questions

Is "which scale for which character" the right question? If you would distinguish these three situations by other means, what would you actually do?

Does Situation 4 strike you as a real thing a character can be in?

Anything wrong with the test itself, or anything you noticed differing between files other than the scale?

What this is	An extra part of the listening test, given only to music therapists
How long	About 30 minutes, after the main test
What you need	Nothing beyond the main pack
Who is asking	J. McKenney, for a research project on whether the choice of scale in dramatic scoring can be written down as a rule
What you get	The results, and credit if you want it

What a therapist can tell us that a composer cannot

1. Why you are being asked separately

Do the main test first. You listen and answer as everyone else does, and your answers go in with theirs. This part is a different thing and should be done afterwards, because some of it will tell you what the main test is careful not to.

The scales in that test come from a system built to score drama. One of that system's published descriptions told therapists they could track a client's psychological state over time and use the music as a discussion tool in a session. Nobody with clinical training was asked whether that was a good idea before it was written. That text is being withdrawn and replaced, and the replacement is in this pack so you can check that it says what we say it says. The questions below are what should have been asked first.

They are not a consultation on how to do it well. They include whether it should be done at all, and we would rather have that answer than a courteous one.

2. What the system is, and what is wrong with it

You should not be asked to judge something whose limits have been kept from you. All of the following is stated in the project's own theory paper and has been confirmed against the code.

The system takes a nine-part description of a character and turns it into a score. For a script imported through the product, four of those nine parts are not read from the text at all but assigned by calls to the random number generator, and on one path two more, the ones the project calls trauma and entropy, are assigned the same way. Three values a user sets in the interface are discarded before they reach the music. The generator is not seeded, so the same input does not produce the same output twice: press render again and you get a different piece.

No quantity in this theory has ever been measured by anything except the system itself. The large set of worked examples in the project is the system's own reading of the author's written commentary on some plays, so the numbers are not an independent assignment by any person. Two listener studies exist, and

neither can be cited as evidence about anything, because the stimuli came from the unseeded generator and cannot be reproduced. One of the two is a null result.

The listening test exists because the project has never once had a number it did not produce about itself. It does not repair any of the above.

3. How to answer

Answer as the clinician you are, not as a reviewer being fair to a project. You are not being asked to accept that the three situations are real, that a state can be mapped to a scale, or that a generated cue has any business in a session. Every question below can be answered by someone who thinks the whole idea is misconceived, and several are asking for precisely that. Where a question rests on a premise you reject, name the premise rather than grading the question. Where you have no view, say so, since a therapist declining to answer is itself information we do not have.

4. The questions

The three situations. These are Situations 1, 2 and 3 from Part 2 of the main test. Situation 4 is discussed there and not here.

1. **Recognition.** Do the first three situations describe anyone you have worked with, or do they describe characters? Answer for each separately if they differ.
2. **Help or hazard.** If a clinician reads one of those descriptions and a particular client comes to mind, is that recognition useful, or is it the first step of a mistake? If it is a mistake, say what the mistake is.
3. **The one you would refuse.** Is there one of the three you would not use as a description of a living person under any circumstances? Which, and what is wrong with it.
4. **What you would have said instead.** If you had to describe what a person is going through to somebody who was going to make music out of it, would you say anything resembling these? If not, what would you actually say.

The session.

5. **The first five minutes.** A client hears a piece that was generated from a description of their state. Walk us through what happens next in the room, as concretely as you can.
6. **When it is wrong.** The client says the music is not right. The system says it is. Who decides, what do you do, and what has just happened to the session.
7. **When it is right, which may be the harder case.** The client says it is exactly right and is moved by it. Is that a good session or a problem, and what do you do with it.
8. **Authority.** Does music that arrives from a machine carry a different authority in the room than music you chose or played yourself? If so, in which direction, and does that difference help you or work against you.

Harm.

9. **Who should not be in the room with this.** Name the presentations you would rule out, and give the mechanism rather than the category: what goes wrong, to whom, how quickly, and whether you would see it happening.
10. **Harm to the work rather than to the client.** Is there a way this damages the relationship, your own attention, or the client's trust in you, without harming the client directly.
11. **The sound itself.** Before any question of meaning, what would have to be true of the audio, including level, onset, predictability and how a client stops it, before you would play a generated piece to anyone.

Consent.

12. **Consent when the output is a claim about the person.** What would a client have to be told, in what words, and at what point.
13. **Disclosing how the numbers arise.** Would you tell a client that some of the values behind their piece were assigned at random, and that the same description produces a different piece each time? If you would, is there anything left to consent to? If you would not, say why not.
14. **Where it goes afterwards.** The state values and the audio are a record of something. Who holds them, who may see them, and what would you refuse to let happen to them.

Supervision and your own standing.

15. **A reading of the client, in the room.** Is "the system's reading of this client" a thing you would ever want present in a session? If it is, under what supervision, how often, and who else has to be able to see it. If it is not, say so plainly, and say whether anything would change your mind.
16. **Your position.** Does using something like this touch your scope of practice, your registration, your insurance, or what your professional body would say to you. We would rather learn this from you now than discover it later.

Before anyone tries it, and whether anyone should.

17. **The thing you would need to see.** Name the specific demonstration or the specific result, not a level of evidence, that would have to exist before you would try this with one person.
18. **The smallest first use.** If there is a version of this small enough to be safe, describe it. If there is not, say there is not.
19. **Where it goes wrong.** If the idea is wrong, we want the first false step rather than a verdict. Point to the place: the description of the state, the assignment of a state to a person, the generation, the playing of it to the person, the interpretation afterwards, or something earlier than all of these.
20. **The other direction.** There is a version in which nobody is read by anything: the client moves the controls until the music matches how they feel, and that piece is what the session works with. Is that a different proposition or the same one with better manners? Answer without assuming we prefer either.
21. **What we did not ask.** What should have been on this list and is not.

5. What happens to your answers

They go into a research paper as data, with the design and the analysis published alongside, including whatever they turn out to say. If the therapists who answer say the clinical application should not exist, the paper says so and the application stops; that is a real outcome of asking and not a formality. You can be named, named as a professional of a stated kind without your name, or anonymous. Say which.